

Crypto Quant Research Report

This report summarizes a systematic crypto trading research project using the TWSQ Trading Library. The project evaluates momentum, breadth-weighted momentum, capitulation-reversal strategies, and combined portfolios under realistic transaction costs.

Research Questions

1. Can cross-sectional crypto momentum outperform BTC buy-and-hold?
2. Can a low-correlation capitulation strategy improve portfolio Sharpe?
3. Are strategy returns explained by BTC beta or by independent alpha?

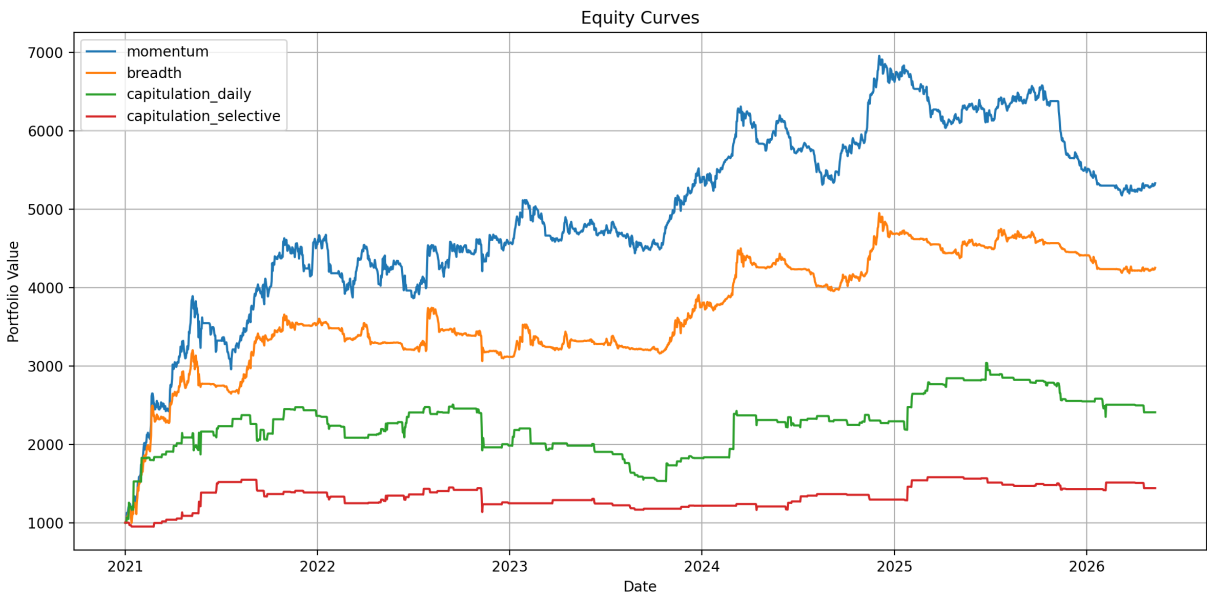
Strategies Evaluated

- 1. Momentum Strategy:** Selects the most liquid cryptocurrencies and ranks them using a volatility-adjusted moving average signal. The strategy buys assets with positive trend scores.
- 2. Breadth-Weighted Momentum:** Extends the momentum strategy by scaling exposure according to market breadth. If many coins are in positive trends, the strategy invests more capital; if market breadth is weak, it reduces exposure.
- 3. Capitulation Bounce Daily:** A reversal strategy that buys liquid assets after large negative daily returns combined with abnormal volume spikes, expecting short-term panic reversals.
- 4. Capitulation Bounce Selective:** A more conservative version of the capitulation strategy with stricter entry filters and lower exposure. It trades less frequently and aims to reduce drawdowns.

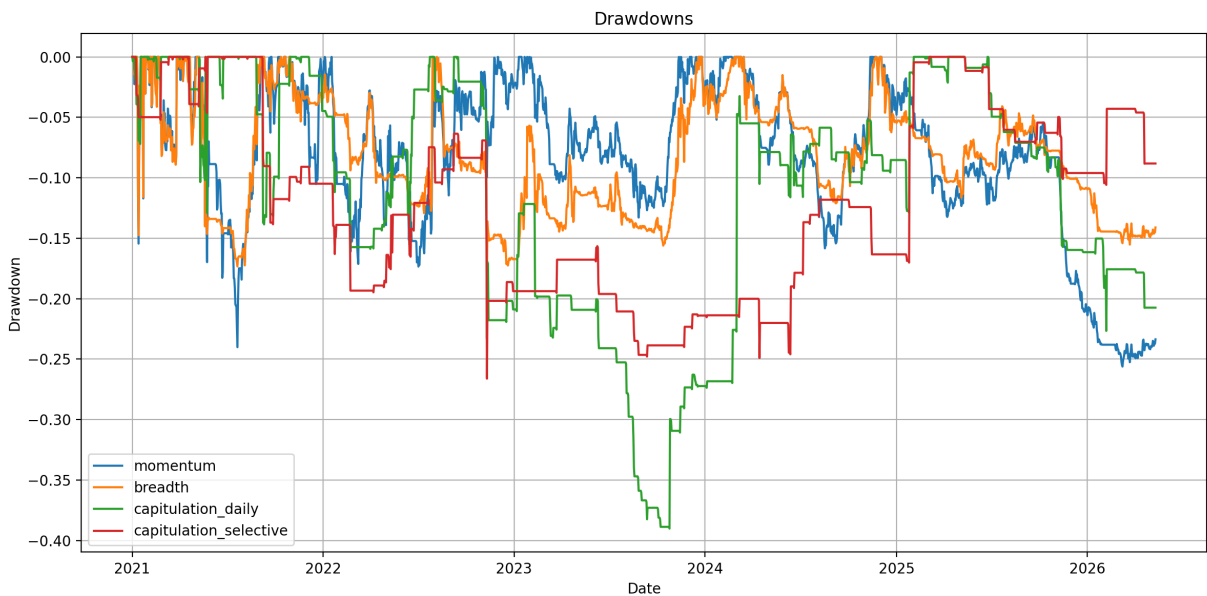
Standalone Strategy Summary

strategy	final_value	profit	total_return	sharpe	max_drawdown	turnover_multiple	num_orders
momentum	5329.4335	4329.4335	4.3487	1.2840	-0.2562	452.5186	5377.0000
breadth	4250.2617	3250.2617	3.2595	1.2620	-0.1819	275.1921	5377.0000
capitulation_daily	2407.3115	1407.3115	1.4073	0.6733	-0.3902	395.4043	664.0000
capitulation_selective	1439.9659	439.9659	0.4400	0.4610	-0.2662	114.5814	270.0000

Equity Curves



Drawdowns



Strategy Correlations

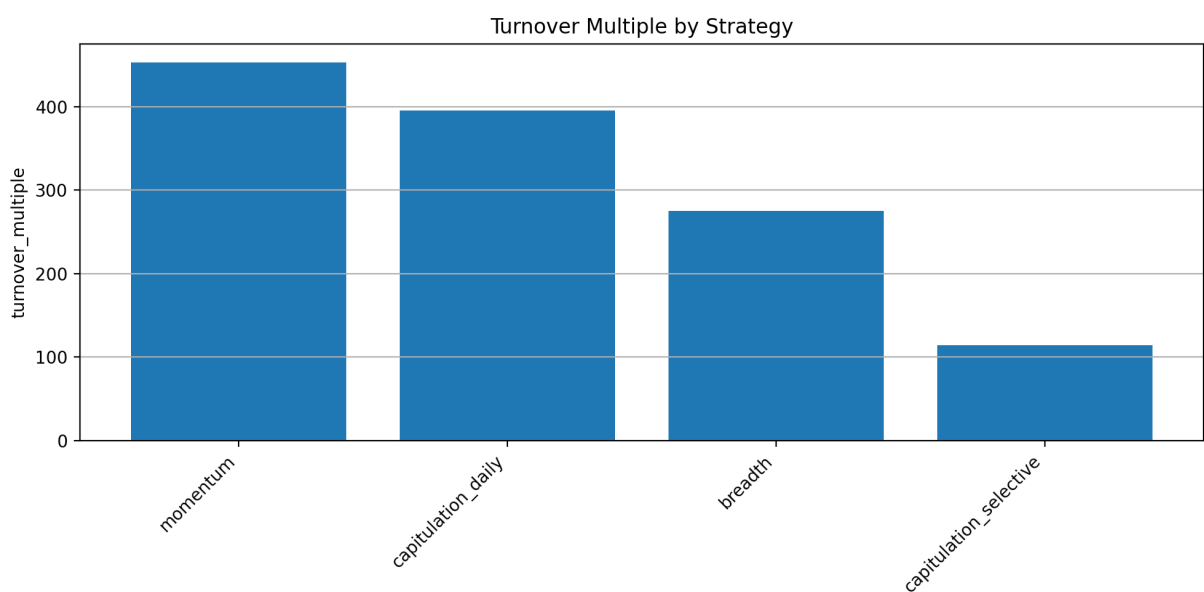
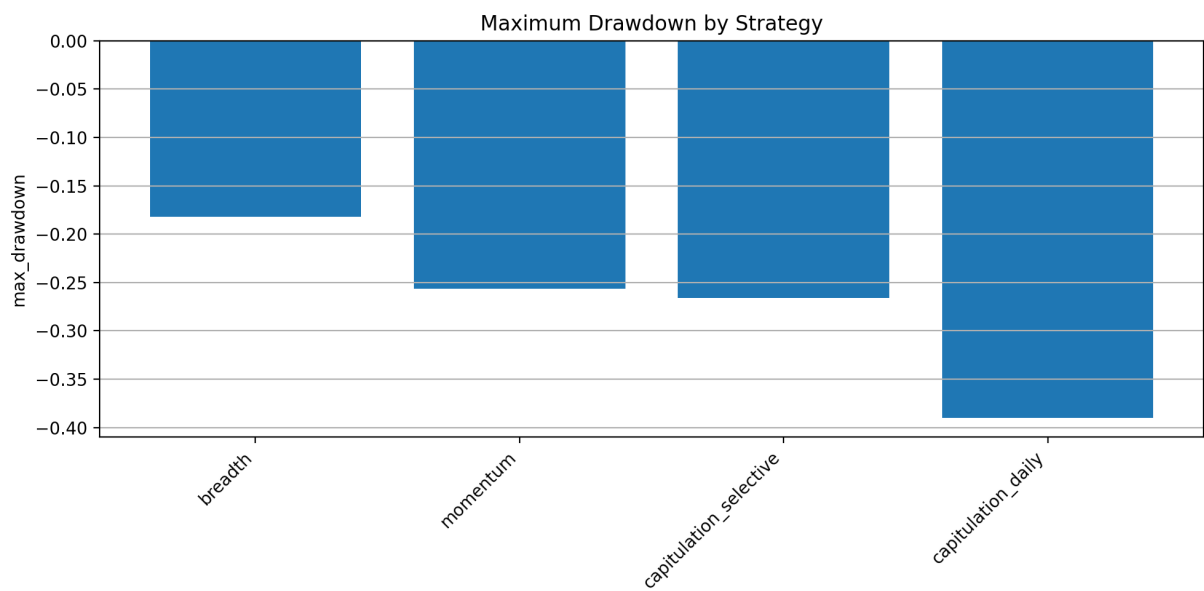
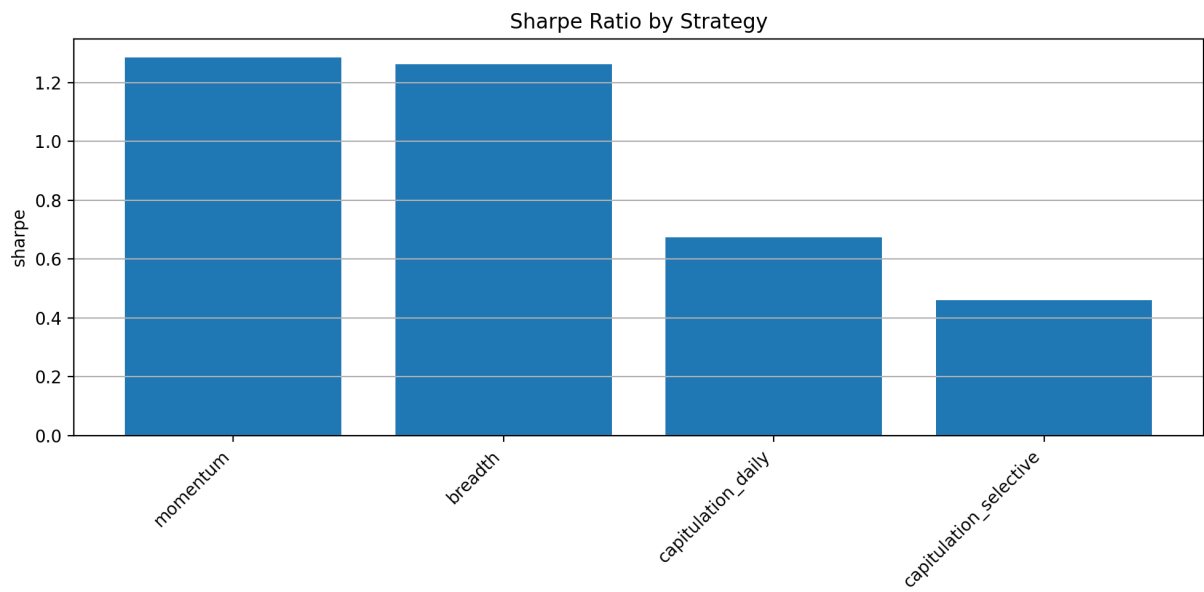
Low correlation between momentum and capitulation strategies suggests that capitulation can act as a diversifying alpha source.

Correlation Matrix

Unnamed: 0	momentum	breadth	capitulation_daily	capitulation_selective
momentum	1.0000	0.8886	0.2697	0.2064
breadth	0.8886	1.0000	0.2394	0.1652
capitulation_daily	0.2697	0.2394	1.0000	0.5506
capitulation_selective	0.2064	0.1652	0.5506	1.0000



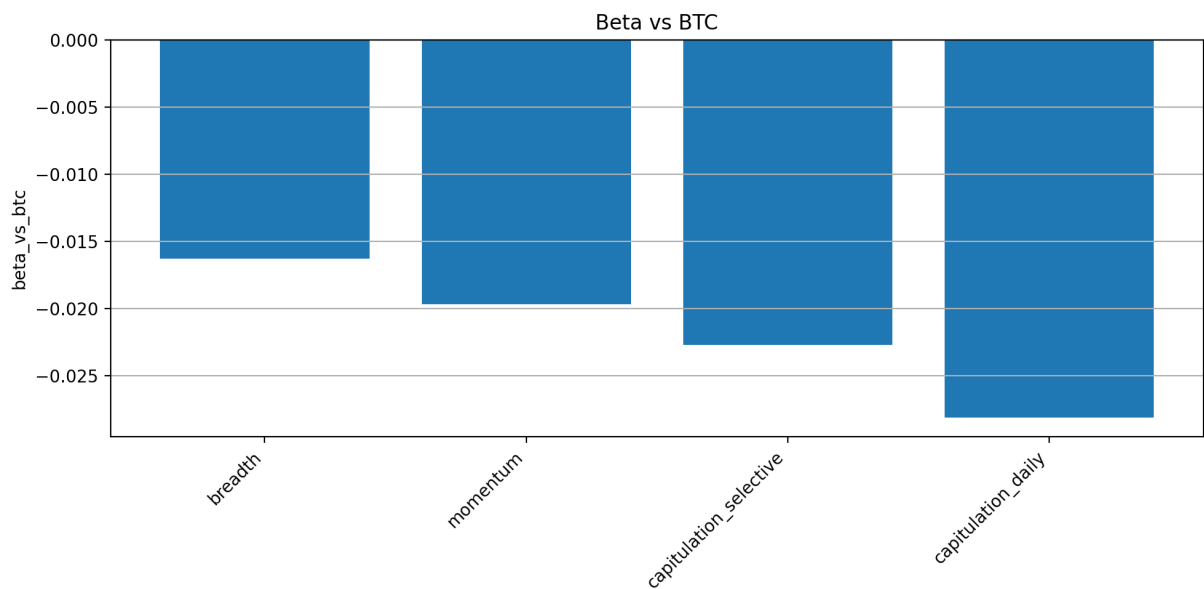
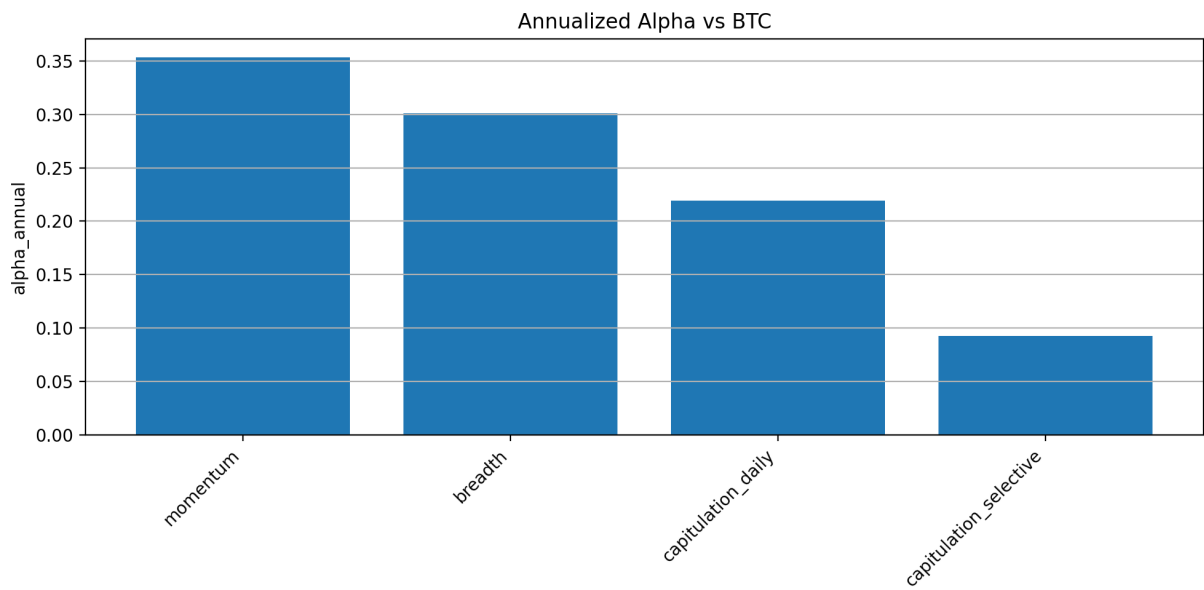
Performance Metrics



Alpha/Beta Regression vs BTC

strategy	alpha_daily	alpha_annual	beta_vs_btc	r_squared	information_ratio	alpha_t_stat	alpha_p_value
momentum	0.0010	0.3532	-0.0197	0.0018	1.2989	3.0031	0.0027
breadth	0.0008	0.3013	-0.0163	0.0016	1.2777	2.9541	0.0032
capitulation_daily	0.0006	0.2190	-0.0281	0.0028	0.7081	1.6371	0.1018
capitulation_selective	0.0003	0.0925	-0.0227	0.0052	0.5051	1.1678	0.2430

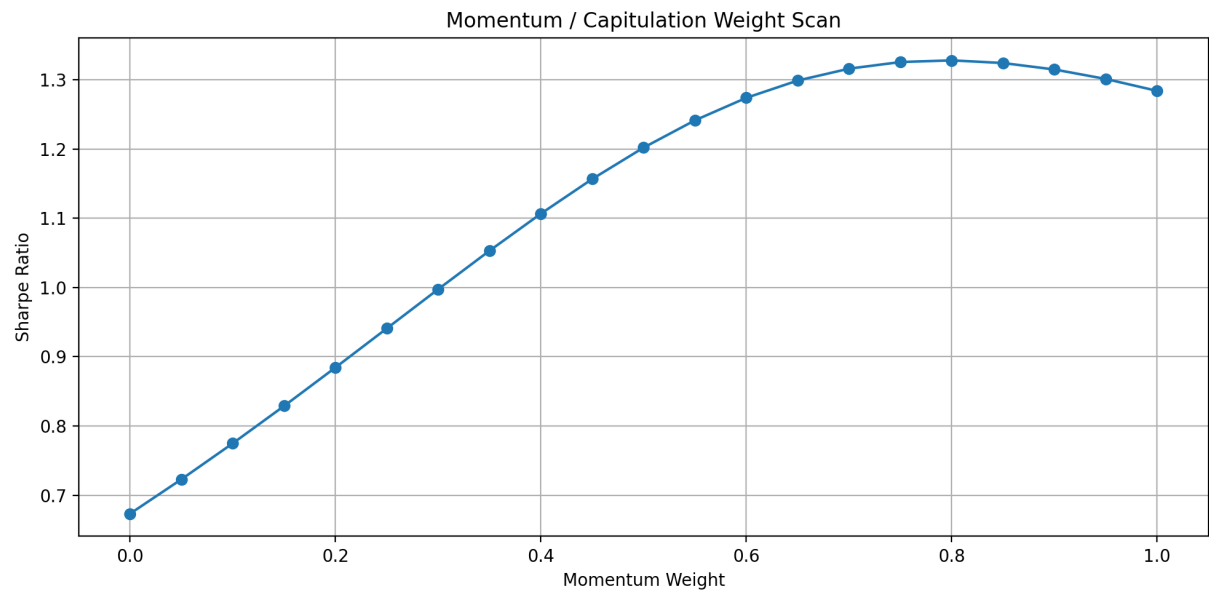
The alpha/beta regression estimates whether the strategy returns are explained by BTC buy-and-hold exposure. Low beta and low R-squared indicate that strategy returns are not simply crypto market beta.



Best Strategy Combinations

left_strategy	right_strategy	left_weight	right_weight	final_value	profit	sharpe	max_drawdown
momentum	capitulation_daily	0.8000	0.2000	4792.7776	3792.7776	1.3281	-0.1941
momentum	capitulation_selective	0.8000	0.2000	4265.7458	3265.7458	1.2995	-0.1850
breadth	capitulation_daily	0.8000	0.2000	3980.2393	2980.2393	1.3186	-0.2028
breadth	capitulation_selective	0.8000	0.2000	3536.6466	2536.6466	1.2875	-0.1850

Momentum + Capitulation Weight Scan



Best Combined Portfolio

The final portfolio combines the strongest standalone alpha, momentum, with the lower-correlation capitulation strategy. The optimal weight was selected by scanning portfolio weights from 0% to 100% in 5% increments and choosing the allocation with the highest Sharpe ratio. The best result was an 80% allocation to momentum and a 20% allocation to daily capitulation.

Best Combined Portfolio Metrics

weights	final_value	profit	sharpe	max_drawdown
{'momentum': np.float64(0.8), 'capitulation_daily': np.float64(0.19999999999999996)}	4792.7776	3792.7776	1.3281	-0.1941

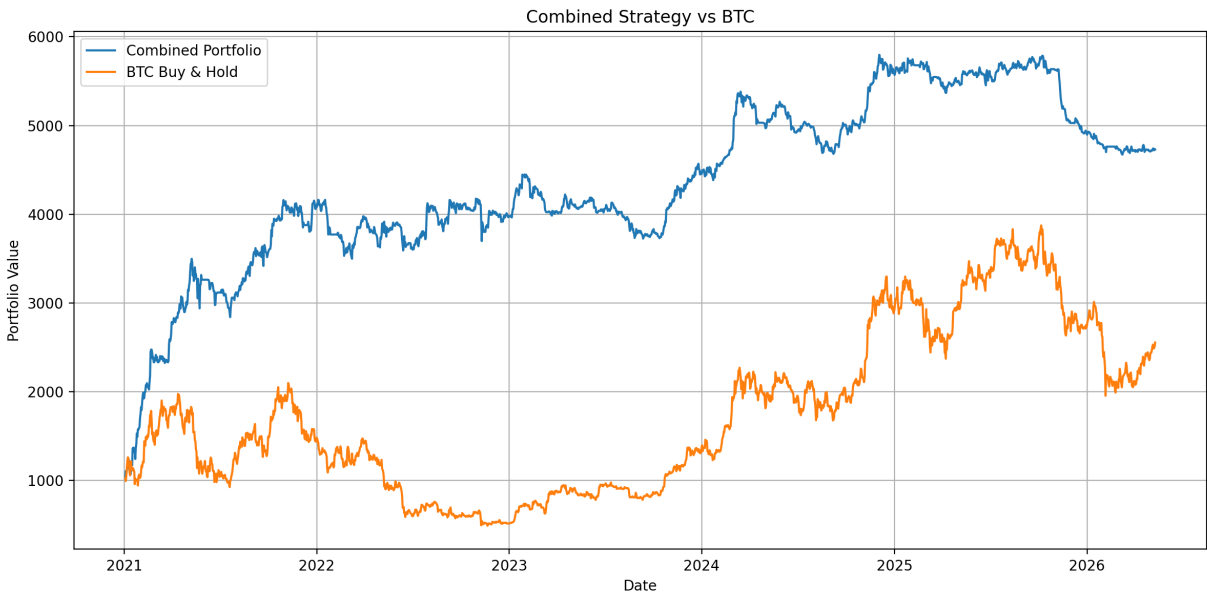
Best Combined Portfolio Alpha Report

alpha_daily	alpha_annual	beta	r_squared	information_ratio	t_stat_alpha	p_value_alpha
0.0009	0.3263	-0.0214	0.0026	1.3509	3.1233	0.0018

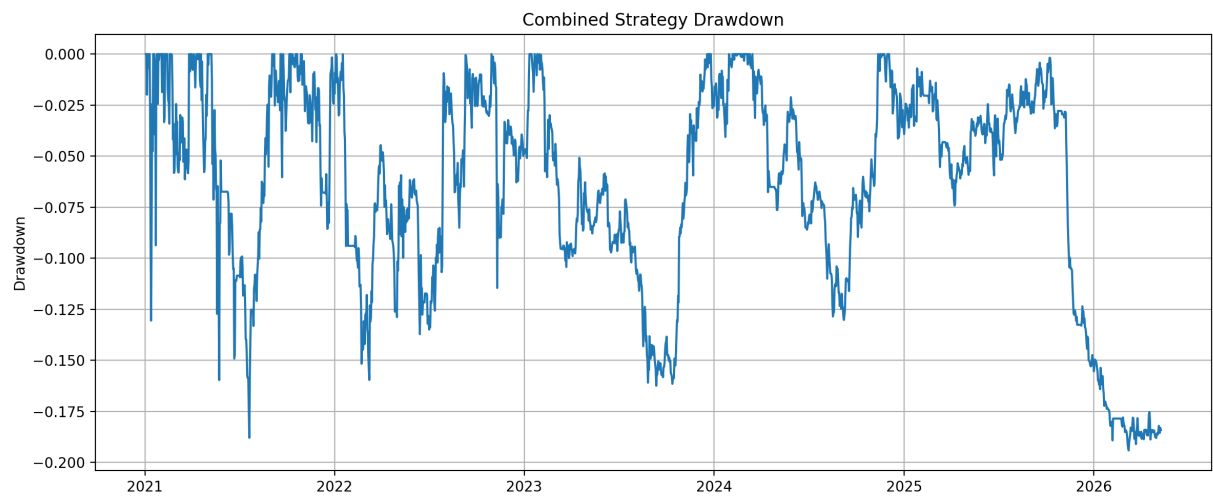
Best Combined Portfolio Benchmark Summary

asset	total_return	annual_return	annual_volatility	sharpe	max_drawdown
strategy	3.7296	0.3368	0.2419	1.3187	-0.1941
benchmark	1.5548	0.1915	0.5809	0.5918	-0.7663

Best Combined Portfolio vs BTC



Best Combined Portfolio Drawdown



Key Findings

The momentum strategy delivered the strongest standalone performance, while the capitulation strategy showed materially lower correlation to momentum. Combining the two improved the portfolio Sharpe ratio and reduced drawdown relative to the standalone momentum strategy.

Limitations

This is a research backtest, not a production trading system. Limitations include possible survivorship bias in the candidate universe, simplified execution assumptions, absence of order book data, no funding-rate data, and limited out-of-sample validation.

Future Work

Future improvements include walk-forward validation, parameter sensitivity analysis, transaction-cost stress tests, volatility targeting, funding-rate integration, and live paper trading.